

## **Reflection: Power Cell Research**

*By Joel Tchouke*

As an aspiring computer engineer, I have always been fascinated by the intersection of technology and problem-solving. Throughout my studies, I learned theoretical concepts that sparked my curiosity, but I often wondered how they could be applied in practical, real-world situations. That curiosity drove me to seek out research opportunities where I could use my knowledge in a tangible way. When I received an email from Professor Winstead looking for students to work on improving the power efficiency of a LEGO brick used in robotics, I knew this was my chance to bridge the gap between theory and practice in hardware design. What followed was a challenging yet rewarding journey that not only enhanced my technical skills but also deepened my understanding of teamwork, problem-solving, and the engineering process.

I first learned about the research opportunity with Professor Winstead through an email he sent to the Department of Electrical and Computer Engineering Technology. He was seeking students interested in working on a project aimed at improving the power efficiency of a LEGO brick used in robotics. The moment I read the email, I knew this was the chance I had been looking for. As a student eager to apply my theoretical knowledge to real-world problems, this project seemed like the perfect fit. I wanted hands-on experience that would challenge me to think creatively and push me out of my comfort zone. My main motivation was to gain research experience and contribute to an innovative project that had the potential to impact robotics. The idea of enhancing a toy's power efficiency while learning about advanced technologies intrigued me, so I quickly reached out to Professor Winstead to express my interest.

Working with Professor Winstead and my research partner was critical to the success of the project. From the beginning, Professor Winstead was both supportive and challenging. He guided us when we needed direction but also encouraged us to explore solutions independently. His deep knowledge of electrical engineering helped us stay on course, and his enthusiasm for the project made me even more excited to dive into the work.

In addition to Professor Winstead's mentorship, I worked closely with my research partner. We developed a strong working relationship by constantly collaborating on solutions, testing ideas, and supporting each other through the tough moments. My partner brought a methodical approach, while I leaned into creative problem-solving, and together we were able to tackle obstacles more effectively. Our collaboration was key to making steady progress, and I learned a lot about teamwork throughout the process.

One of the biggest challenges we faced early on was selecting the right battery for the LEGO brick. The battery needed to be small enough to fit inside the brick but powerful enough to provide a long-lasting, rechargeable solution. We went through several rounds of testing different batteries, running into failures when they either didn't fit or couldn't hold enough charge. Eventually, we found a 1300mAh rechargeable battery that met both the size and capacity requirements. While this felt like a significant breakthrough, it was only the beginning.

Once we had the battery, we turned our attention to designing the circuitry. Here, space was the biggest constraint—we needed to design a compact circuit that could fit inside the tiny LEGO brick, all while including essential features like a battery level indicator and a sleep mode to conserve power. To achieve this, we selected the ATtiny85 microcontroller as the brain of the circuit. Its small size and low power consumption made it the perfect fit for our compact design. Using the ATtiny85 allowed us to control the various components and functions efficiently while keeping the power usage low.

To design the PCB, we used KiCad, an open-source PCB design software. It helped us create compact and efficient circuit boards that could fit inside the LEGO brick. Designing a functional circuit with limited space was a steep learning curve, especially since neither my partner nor I had much experience with PCB design. But through experimentation and multiple prototypes, we were able to develop a circuit that met the project's requirements.

In addition to PCB design, we also used MATLAB to simulate power usage and test how different circuit configurations would affect the battery's efficiency. MATLAB allowed us to visualize and predict how our circuit would perform under different conditions, helping us fine-tune our design before moving on to physical prototypes. This combination of software tools—KiCad for physical design and MATLAB for simulation—helped us overcome many of the technical limitations we faced.

The key to overcoming the challenges we faced was a combination of persistence and collaboration. When we struggled to find the right battery, we conducted extensive research, tested different options, and analyzed our failures. After finding the 1300mAh battery, we shifted focus to the more intricate problem of designing the circuitry. One of our problems was raising the voltage of the battery to 5V as we could only get a battery providing 3.7V. This was another problem for which we did not yet have the skills for. We explored many options, ranging from the use of transistors and operational amplifiers to integrated chips that boosted voltage. None of those solutions seemed viable. We then turned to Professor Winstead, who advised us to build our own circuit to boost the voltage using what we call an “RLC” circuit. That gave me hope because it reminded me that when I am stuck, I can rely on guidance from more experienced engineers.

Designing a compact circuit required a methodical approach. We used KiCad to create several iterations of our PCB layout, testing each design for size, functionality, and efficiency. There were moments when things didn't work as planned; components didn't fit, connections failed, or the circuit didn't perform as expected. By continually refining our designs and using MATLAB to simulate different scenarios, we gradually found a solution. The ability to test and iterate using these tools allowed us to tackle problems early in the design process and reduce issues when creating physical prototypes.

The trial-and-error process I experienced in this project taught me invaluable lessons about perseverance and problem-solving. Despite all the time and careful planning I invested, things didn't always go as expected—a reminder that setbacks are an inevitable part of meaningful work. However, I realized that as long as I didn't give up, I could eventually find a way forward. There

were moments when I felt stuck and wanted to quit because the problems seemed insurmountable. Yet, with Professor Winstead's encouragement and guidance, I found the strength to keep going. This project showed me that both life and engineering challenges are rarely meant to be tackled alone; with the right people, even the most difficult goals become achievable.

One of the most rewarding aspects of the project was successfully designing a sleep mode for the LEGO brick. When not in use, the brick would enter a low-power state, conserving energy and extending battery life. We also created a battery level indicator that allowed users to easily check how much power remained. These features made the LEGO brick more user-friendly and energy-efficient, aligning perfectly with the project's goals.

In the context of our power cell project, ethical use of information was essential at every stage, from initial research to final testing. Working with battery and power efficiency data required ensuring that measurements and results were accurate and transparently reported, as these factors directly impact safety and functionality in real-world robotics applications. Any inaccuracy could lead to overestimations about battery life or power stability, which could compromise safety and reliability.

Furthermore, while designing the battery system, we considered environmental sustainability. Power efficiency is not only about improving performance; it is also about reducing waste and conserving resources. By prioritizing these ethical considerations, we aimed to create a system that performs well while remaining sustainable and safe, reinforcing trust in engineering work.

Looking back on this research project, I am struck by the depth of growth I experienced as an engineer. Beyond technical skills like PCB design and power management, I learned how to translate theoretical knowledge into practical solutions. Gaining proficiency in KiCad, MATLAB, and ATtiny85 microcontroller integration expanded my technical toolkit and strengthened my confidence in tackling complex engineering challenges.

The hands-on challenges I faced prepared me for future projects by teaching me how to operate within constraints such as size, time, and resources. I learned that engineering is not solely about solving problems but about solving them efficiently and responsibly. This mindset continues to influence how I approach technical challenges today.

The teamwork aspect of the project was equally important. Collaborating with my research partner and learning from Professor Winstead taught me how to communicate clearly, listen to alternative ideas, and combine different problem-solving approaches. These interpersonal skills are essential in engineering, where collaboration often determines project success.

One of the most impactful lessons I gained from this project is that failure is not only inevitable in engineering but instrumental to learning. Each setback pushed me to think more critically and creatively. I learned to embrace failure as an opportunity for growth rather than a dead-end.

I do have one regret from this project. We chose not to present our research publicly because I believed it was too small or not innovative enough for a symposium. Looking back, I recognize

that decision came from lack of confidence. In the future, I will share my research regardless of scale because knowledge grows when it is shared.

The skills and lessons I gained from this project will be valuable both professionally and personally. Learning PCB design, circuit simulation, and embedded system control has prepared me to handle complex engineering challenges. The teamwork and communication skills I developed will help me succeed in collaborative environments. Personally, I became more resilient and adaptable, learning that persistence and collaboration are essential to overcoming challenges. This project immersed me in the practical realities of engineering and strengthened my commitment to continuous learning and innovation.